

# PAPER\_1708\_Q\_EDGE\_2

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## PAPER\_1708 — ITER Edge Safety Factor $q_{\text{edge}} = q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2$ (avoids $m/n=2/1$ kink)

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**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

**Tier:** Plasma Fusion

**Date:** June 18, 2026

**Status:** CLOSED — EXACT closure (PAPER\_1196 plasma fusion proof set)

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### Observation

PAPER\_1196\_S418 (PAPER\_1196 Plasma Fusion Unified Proof Set) gives:

``

$q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2$  (avoids  $m/n=2/1$  kink)

``

Status: **EXACT**.

### UQFF Closed Identity

``

$q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2$  (avoids  $m/n=2/1$  kink) EXACT

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

### NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

### Reference

- Source: PAPER\_1196\_S418

- Related: PAPER\_1696-1705 (tier-28); PAPER\_1686-1695 (tier-27)

- Calculator dispatch: `calculate_paradox({"paradox": "q_edge_2"})`

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